junction kept in a sealed plastic or glass casing. The cover is so designed that the light rays are allowed to fall on one surface across the junction. The remaining sides of the casing are painted to restrict the penetration of the light rays. A lens permits light to fall on the junction.

Working Principle: When light falls on the reverse-biased PN photo diode junction, electron-hole pairs are created. The movement of these electron-hole pairs in a properly connected circuit results in current flow. On illumination on the diode, the magnitude of the photo current depends on the number of charge carries generated. This current is also affected by the frequency of the light falling on the junction of the photo diode.

Applications

1. Photo diodes are used as light-detectors, demodulators and encoders.
2. They are also used in optical communication system.
3. They are used in high speed counting and switching circuits.

6.6. PHOTO TRANSISTOR

Photo transistor is a much more sensitive semiconductor photo device than the PN photodiode. The current produced by a photodiode is very low which cannot be directly used in control applications. Therefore, this current should be amplified before applying to the control circuits. The photo transistor is a light detector which combines a photodiode and a transistor amplifier. When the photo transistor is illuminated, it permits the flow of current.

History: Phototransistor was made by John N. Shive and his team members from Bell Laboratory during the years 1948-1950.



Figure 6.6 shows the NPN phototransistor. It is usually connected in common emitter (CE) configuration with base open. A lens focuses the light on the base-collector junction. The modern photo transistor uses efficient light effective materials instead of making a hole and fixing a lens on it. Photo transistor has three terminals, but only two terminals are generally used. Here the base current is supplied by the current created by the light falling on the base-collector junction.

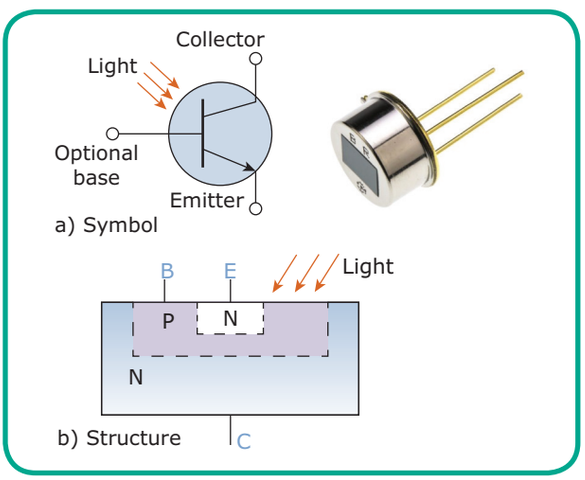


Figure 6.6 NPN Photo Transistor

The voltage (V_{CE}) applied to the transistor makes emitter-base junction forward biased and collector-base junction reverse

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